



KC-7442

**Integrated M. C. A. (Sem. V)
Examination - December - 2024**

~~IMCA - 502T~~

Data Science

Time allowed : Three Hours

Max. Marks : 70

Min. Marks : 28

*This question paper contains two sections
as under:*

Section-A**Max. Marks-20**

This section will have 10 questions, having 02 questions from each unit with no internal choice. Answer in about 20 words. All question will have equal marks. (10×02=20)

Section-B**Max. Marks-50**

This section will have 10 questions, having 02 questions from each unit, one question to be attempted from each unit. Words limit approximately 400-500 words. Questions may have sub divisions. All question will have equal marks. (10×05=50)

SECTION - A

- 1 (i) What is Data Science ? What are its application ?
- (ii) What do you mean by Outlier ?
- (iii) What is data modelling ?
- (iv) Differentiate Univariate and Bivariate.
- (v) How can you avoid overfitting in a model ?
- (vi) What is Training data and Testing data ?
- (vii) What is confusion matrix ?
- (viii) Differentiate between supervised and unsupervised learning.
- (ix) Name any 4 classification algorithms.
- (x) What do you mean by best fit and worst fit.

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[Contd...

SECTION - B**UNIT - I**

- 2 Explain all the aspects of machine learning pipeline.

OR

How to clean and handle missing data in a dataset ? Explain.

UNIT - II

- 3 Explain the exploratory data analysis.

OR

What is feature engineering ? What is data modelling ?

UNIT - III

- 4 What is overfitting and underfitting in Machine Learning ? Explain Goodness of fit.

OR

Explain Bias and Variance in Machine Learning. How Bias-variance Trade-off is done in ML ?

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[Contd...

UNIT - IV

- 5** Explain linear and logistic regression with regularization.

OR

✓ What is support vector machine ? Explain with example.

UNIT - V

- 6** ✓ Explain the role of Python in Data Science.
What are key libraries used, and how do they help in data analysis ?

OR

What is Numpy ? Explain its key features and demonstrate how to create and manipulate Numpy arrays with example.
